

# Assignment 3: Raw Socket Packet Analysis and Protocol Investigation

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## 1. Objective

The objective of this assignment is to understand raw socket programming, IP packet structure, protocol headers, and packet analysis using Linux and Wireshark. Students will write a C program using raw sockets, generate protocol-specific traffic, capture packets, compare results with Wireshark, and analyze packet header fields .

## 2. System configuration

***Operating system (uname -a )*** : Linux msi UBUNTU

***Compiler (gcc --version )*** : gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0

***Network (ip addr) :***

1: lo: <LOOPBACK,UP,LOWER\_UP> mtu 65536 qdisc noqueue state UNKNOWN  
group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

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```
valid_lft forever preferred_lft forever

inet 10.255.255.254/32 brd 10.255.255.254 scope global lo

valid_lft forever preferred_lft forever

inet6 ::1/128 scope host proto kernel_lo

valid_lft forever preferred_lft forever

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP
group default qlen 1000

link/ether 00:15:5d:39:22:8f brd ff:ff:ff:ff:ff:ff

altname enx00155d39228f

inet 192.168.67.131/20 brd 192.168.79.255 scope global eth0

valid_lft forever preferred_lft forever

inet6 fe80::215:5dff:fe39:228f/64 scope link proto kernel_ll

valid_lft forever preferred_lft forever
```

### 3. Compile and Run Commands

```
gcc raw_capture.c -o raw_capture

sudo ./raw_capture
```

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## 4. Program Design

The program was implemented using Linux raw sockets with protocol ICMP.

Assigned Protocol : **ICMP**

Protocol Number : **1**

Information Displayed

IP Header Fields

- Source IP Address
- Destination IP Address
- Protocol Number
- TTL Value
- Packet Size

Protocol Specific Fields

- ICMP Type
- ICMP Code

The program captured and displayed information for 20 ICMP packets.

## 5. Traffic Generation

Command Used : **ping -c 20 8.8.8.8**

Interface used : given below in .png

Program output : In program output.txt

```
piyush@MSI:/mnt/c/Users/harsh/OneDrive-piyush/OneDrive/Desktop/ISEA/CSB24051_PIYUSH_KUMAR_RAWSOCKET$ ping -c 20 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.048 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.044 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.507 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.043 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.403 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.062 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.301 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.267 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.174 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.034 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.035 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.358 ms
64 bytes from 127.0.0.1: icmp_seq=15 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=16 ttl=64 time=0.107 ms
64 bytes from 127.0.0.1: icmp_seq=17 ttl=64 time=0.044 ms
64 bytes from 127.0.0.1: icmp_seq=18 ttl=64 time=0.032 ms
64 bytes from 127.0.0.1: icmp_seq=19 ttl=64 time=0.046 ms
64 bytes from 127.0.0.1: icmp_seq=20 ttl=64 time=1.30 ms

--- 127.0.0.1 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19459ms
rtt min/avg/max/mdev = 0.032/0.198/1.296/0.289 ms
```

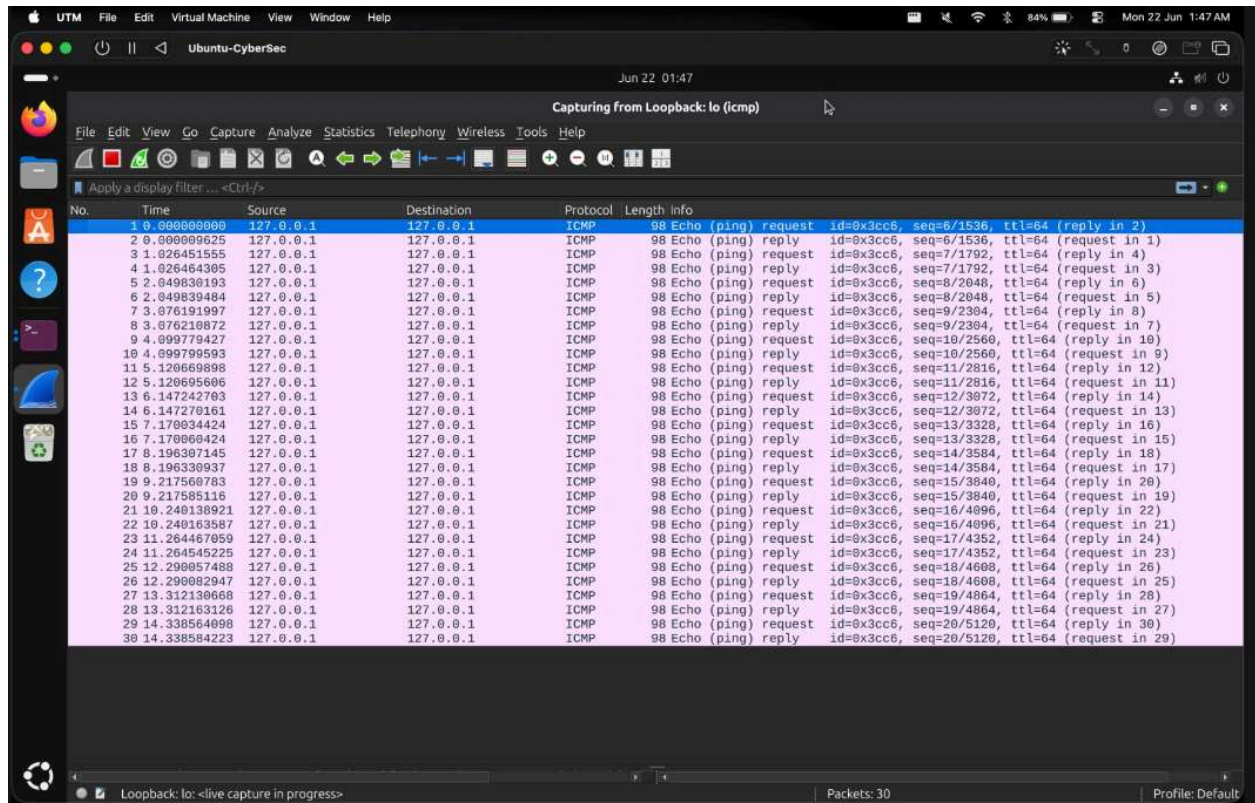
## 6. Experimental Results

```
ROLL_NO=CSB24051
ASSIGNED_PROTOCOL=ICMP

PACKET_NO=1
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=ICMP
PROTOCOL_NO=1
TTL=64
PACKET_SIZE=84
ICMP_TYPE=8
ICMP_CODE=0
IP_VERSION=4
```

Similarly 20 packets were captured and saved in program\_output.txt

## 7. Wireshark Verification



## 8. Header Analysis

### 1. Why is TTL not zero?

TTL prevents packets from circulating indefinitely in the network. Every router decrements the TTL value by one. If it becomes zero, the packet is discarded.

### 2. Why do packet sizes differ?

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Packet sizes vary due to differences in payload size and header information.

### **3. What information does the protocol field provide?**

The protocol field identifies the upper-layer protocol carried by the IP packet.

Examples:

- ICMP = 1
- TCP = 6
- UDP = 17

### **4. What happens if the protocol field is modified?**

The receiver may interpret the packet incorrectly or discard the packet.

### **5. What does the protocol-specific field reveal?**

ICMP Type and Code indicate the purpose of the packet.

Observed values:

ICMP Type = 0

ICMP Code = 0

which indicate ICMP Echo Reply packets.

## **9. Program Enhancement**

### **Selected Field: IP Version**

#### **1. Purpose of the Selected Field**

The IP Version field indicates the version of the Internet Protocol used by the packet. It helps network devices interpret the packet format correctly. A value of 4 represents IPv4, while a value of 6 represents IPv6.

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## 2. Observed Values from Five Captured Packets

| Packet Number | IP Version |
|---------------|------------|
| 1             | 4          |
| 2             | 4          |
| 3             | 4          |
| 4             | 4          |
| 5             | 4          |

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## 3. Whether the Value Stayed the Same or Changed

The IP Version value remained constant for all captured packets. Every packet had an IP Version value of 4, indicating that all packets captured during the experiment were IPv4 packets. No variation was observed because the communication was carried out using the IPv4 protocol.

## 10. Reflection Answers

### 1. Why are root privileges required for raw sockets?

Raw sockets provide direct access to network packets and headers. Root privileges prevent unauthorized access and packet manipulation.

### 2. How are raw sockets different from TCP/UDP sockets?

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Raw sockets provide low-level access to packets and headers, whereas TCP and UDP sockets provide transport-layer communication with automatic processing.

### **3. One advantage and one limitation of raw sockets**

Advantage: Enables packet analysis and custom protocol implementation.

Limitation: Requires root privileges and is more complex to program.

### **4. One networking or cybersecurity application**

Raw sockets are used in packet sniffers, intrusion detection systems, and network diagnostic tools.

## **11. Conclusion**

A raw socket program was successfully implemented to capture ICMP packets. Twenty packets were analyzed and verified using Wireshark. Various IP header fields and ICMP-specific fields were examined, and the observations obtained from the raw socket program matched the Wireshark capture. The experiment provided practical understanding of packet structure and protocol analysis using raw sockets.